

Fit risoluzione in p_T vs $|\eta^{\text{truth}}|$: r_0 libero vs $r_0 = 0$

$$\sigma_{p_T}/p_T = \sqrt{r_0^2/p_T^2 + r_1^2 + (r_2 \times p_T)^2} \quad -- \quad \sigma_{68} = (q_{84} - q_{16})/2 \quad -- \quad Z + Z. \text{ combinati}$$

$ \eta $ bin	n punti	r_0 libero [GeV]	χ^2/ndf (r_0 libero)	χ^2/ndf ($r_0 = 0$)	r_2 [GeV ⁻¹]
$0.0 \leq \eta < 0.1$	16	0.000 ± 0.003	7590.92	7048.71	$0.214 \pm 0.000 \ (\times 10^{-3})$
$0.1 \leq \eta < 1.05$	16	0.000 ± 0.001	83848.68	77859.48	$0.090 \pm 0.000 \ (\times 10^{-3})$
$1.05 \leq \eta < 1.3$	15	0.000 ± 0.006	6212.69	5734.79	$0.090 \pm 0.000 \ (\times 10^{-3})$
$1.3 \leq \eta < 1.7$	14	0.000 ± 0.003	22828.11	20925.77	$0.142 \pm 0.000 \ (\times 10^{-3})$
$1.7 \leq \eta < 2.5$	13	0.000 ± 0.003	12682.86	11529.87	$0.103 \pm 0.000 \ (\times 10^{-3})$
$2.5 \leq \eta < 2.8$	11	0.322 ± 0.002	81.15	1330.19	$0.112 \pm 0.001 \ (\times 10^{-3})$

In rosso: $\chi^2/\text{ndf} > 3$, fit poco affidabile

In blu: r_2 , la risoluzione nominale ATLAS ad alto p_T